```
> restart: with(plots): with(plottools):with(LinearAlgebra):with
  (VectorCalculus):BasisFormat(false):with(GEO3):
> with(CurveFitting):with(ColorTools):
```

Farver anvendt i projektet

```
> mg:=Color([36, 59, 52])
```

$$mg := \langle \text{RGB} : 0.141\ 0.231\ 0.204 \rangle \quad (1)$$

```
> lg:=Color([136, 202, 156])
```

$$lg := \langle \text{RGB} : 0.533\ 0.792\ 0.612 \rangle \quad (2)$$

Der fastsættes, at alle numeriske resultater vises med seks decimaler efter kommaet.

```
> Digits:= 6;
```

$$Digits := 6 \quad (3)$$

C.1 Konstanter der kan parametiseres

Radius af objektet, når objektet er cirkulært

```
> rad:= 0.5
```

$$rad := 0.5 \quad (1.1)$$

Objektets bredde

```
> a:= 1:'a'=a
```

$$a = 1 \quad (1.2)$$

Objektets højde:

```
> b:= 1:'b'=b
```

$$b = 1 \quad (1.3)$$

Løkkens bredde

```
> c:=0.742:'c'=c
```

$$c = 0.742 \quad (1.4)$$

Løkkens længde

```
> d:= 1.023:'d'=d
```

$$d = 1.023 \quad (1.5)$$

Løkkens placering i x retningen

```
> e:= -0.152:'e'=e
```

$$e = -0.152 \quad (1.6)$$

Løkkens placering i y retningen

```
> f:= 0:'f'=f
```

$$f = 0 \quad (1.7)$$

Løkkens placering i z retningen

```
> g:= -0.996:'g'=g
```

$$g = -0.996 \quad (1.8)$$

"Reb" længde

```
> h:= 2.639:'h'=h
```

$$h = 2.639 \quad (1.9)$$

"Reb" længde

```
> i:= 2.068:'i'=i
```

$$i = 2.068 \quad (1.10)$$

Løkkens hældning

```
> j:= 3:'j'=j
```

$$j = 3 \tag{1.11}$$

C.2 Den første løkke

Der konstrueres vektorfunktioner for fire kvarte cirkler, én halvcirkel samt to vandrette, lineære funktioner.

```
> cirk1:= t-> [(-1+cos(t))*d, (2-sin(t))*c, 0]:'cirk1(t)'=cirk1(t)
```

$$cirk1(t) = [-1.023 + 1.023\cos(t), 1.484 - 0.742\sin(t), 0] \tag{2.1}$$

```
> cirk1Fig:=tubeplot(cirk1(t),radius=rad, t=0...Pi/2,  color=mg,
  scaling=constrained):
```

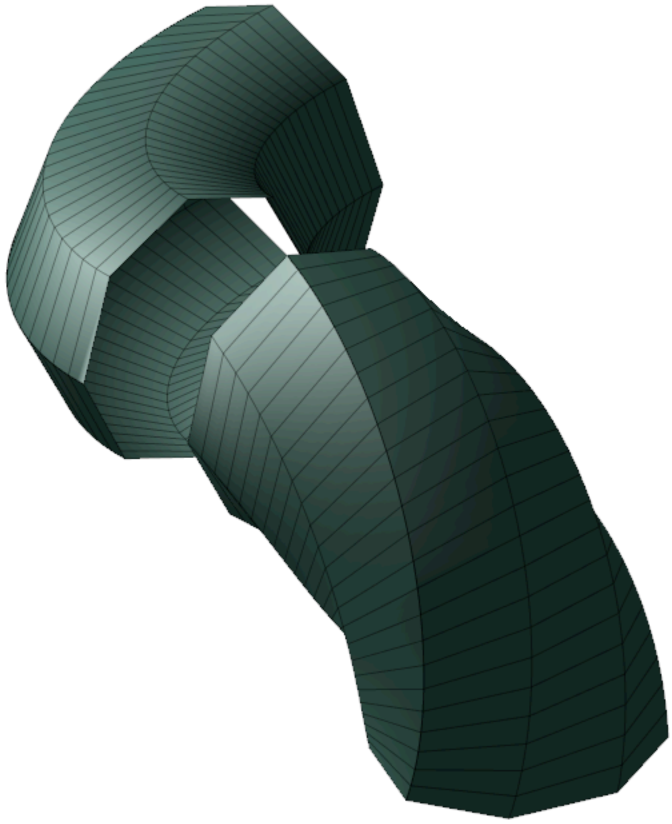
```
> cirk2:= t-> [0, 2+cos(t), sin(t)-1]:'cirk2(t)'=cirk2(t)
```

$$cirk2(t) = [0, 2 + \cos(t), \sin(t) - 1] \tag{2.2}$$

```
> cirk2Fig:=tubeplot(cirk2(t),radius=rad, t=0....Pi/2,  color=mg,
  scaling=constrained):
```

Kurvedelene samles endnu ikke, da der senere tilføjes konstanter, som skal sikre korrekt sammenhæng mellem delene.

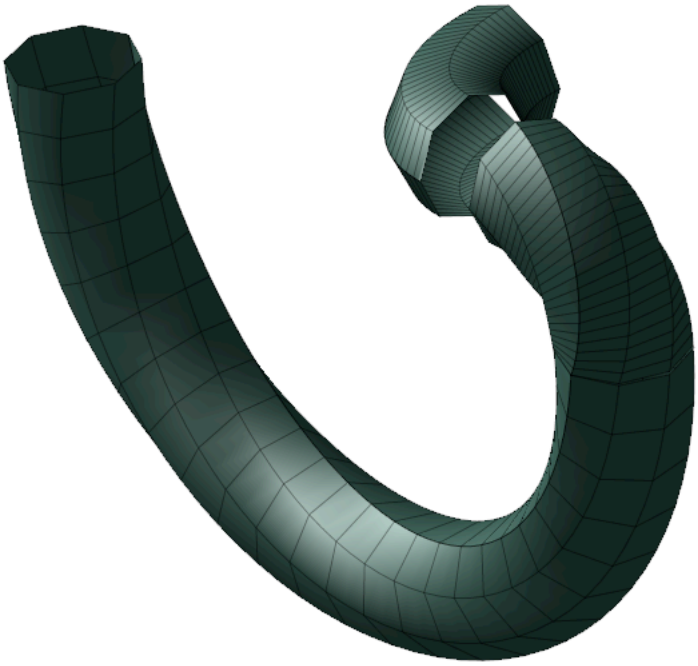
```
> display(cirk1Fig, cirk2Fig, axes=none,orientation=[42,59,-8])
```



```
> cirk3:= t-> [0, 3*cos(t), -3*sin(t)-1]:'cirk3(t)'=cirk3(t)
```

$$cirk3(t) = [0, 3\cos(t), -3\sin(t) - 1] \qquad \textbf{(2.3)}$$

```
> cirk3Fig:= tubeplot(cirk3(t),radius=rad, t=0...Pi,  color=mg,
  scaling=constrained):
> display(cirk1Fig, cirk2Fig, cirk3Fig, axes=none,orientation=[42,
  59,-8]);
```



```
> cirk4:= t->  [0, -2-cos(t), -1+sin(t)]:'cirk4(t)'=cirk4(t)
```

$$cirk4(t) = [0, -2 - \cos(t), \sin(t) - 1] \tag{2.4}$$

```
> cirk4Fig:=tubeplot(cirk4(t),radius=rad, t=0....Pi/2,  color=mg,
  scaling=constrained):
> display(cirk1Fig, cirk2Fig, cirk3Fig, cirk4Fig, axes=none,
  orientation=[42,59,-8]);
```

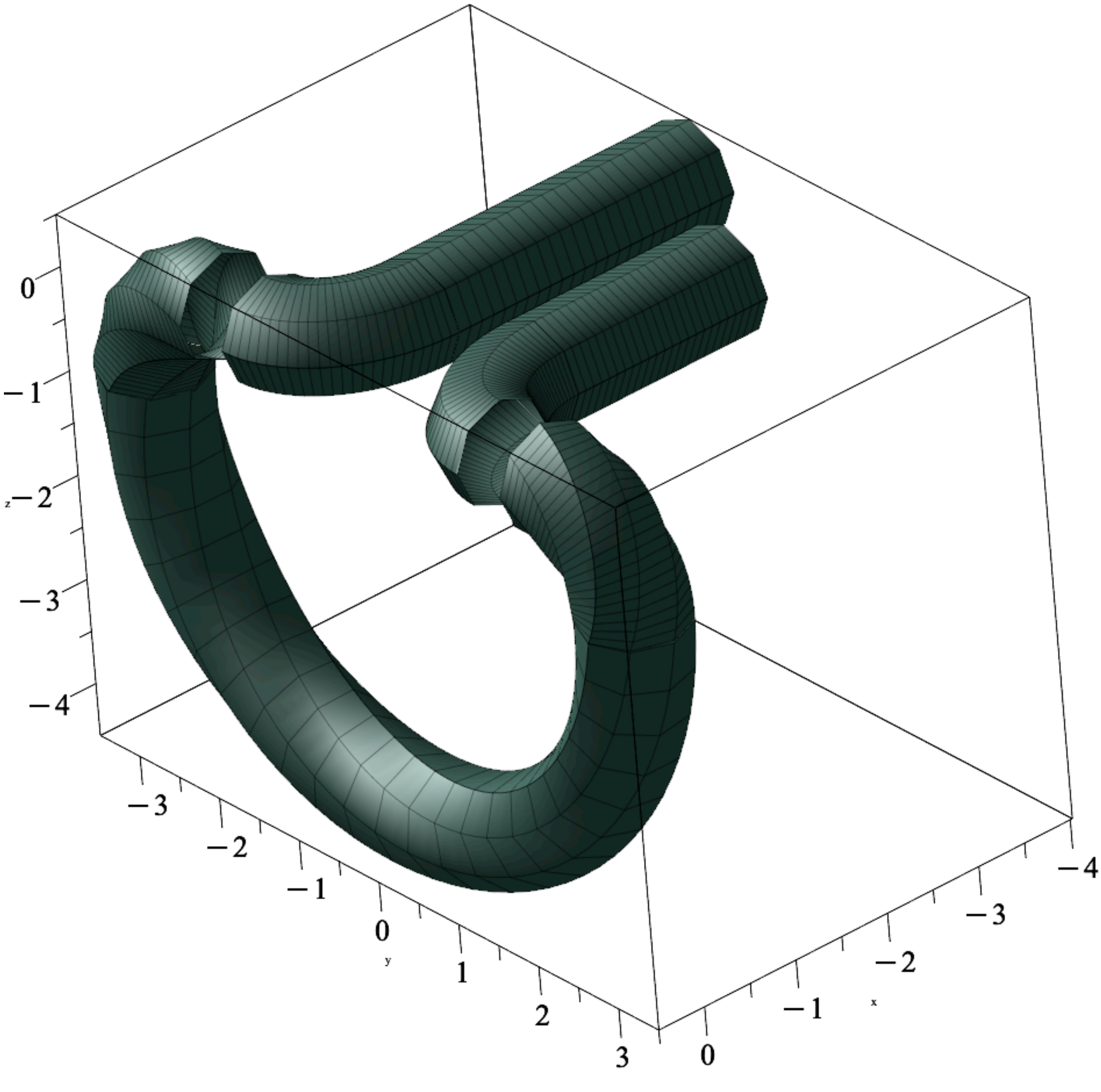
```
> cirk5:= t-> [(-1+cos(t))*d, (-2+sin(t))*c, 0]:'cirk5(t)'=cirk5
  (t)
```

$$cirk5(t) = [-1.023 + 1.023\cos(t), -1.484 + 0.742\sin(t), 0] \quad \textbf{(2.5)}$$

```
> cirk5Fig:= tubeplot(cirk5(t),radius=rad, t=0....Pi/2,  color=mg,
  scaling=constrained):
> display(cirk1Fig, cirk2Fig, cirk3Fig, cirk4Fig, cirk5Fig, axes=
  none,orientation=[42,59,-8]);
```

```
> cyl1:= tubeplot([-d-t, c,0], radius=rad, t=0...i, color=mg,
  scaling=constrained):
```

```
> cyl2:= tubeplot([-t-d, -c,0], radius=rad, t=0...j, color=mg,
  scaling=constrained):
> display(labels=["x","y","z"],cirk1Fig, cirk2Fig, cirk3Fig,
  cirk4Fig, cirk5Fig, cyl1, cyl2,orientation=[42,59,-8]);
```



Rotation af midterste del omkring y-aksen (i positiv x-retning):

C.2.1 Rotation af lodret del af løkken

Rotationen af det lodrette stykke af løkken skal udføres omkring y-aksen. Derfor bestemmes rotationsmatricen om y-aksen:

```
> Ry:= v-> Matrix(3,3, [[cos(v), 0, -sin(v)], [0, 1, 0], [sin(v), 0, cos(v)]]):'Ry(v)'=Ry(v)
```

$$Ry(v) = \begin{bmatrix} \cos(v) & 0 & -\sin(v) \\ 0 & 1 & 0 \\ \sin(v) & 0 & \cos(v) \end{bmatrix} \qquad (2.1.1)$$

Vinklen til rotationen skal angives i radianer og bestemmes på følgende måde:

```
> theta:=Pi/j:'theta'=theta
```

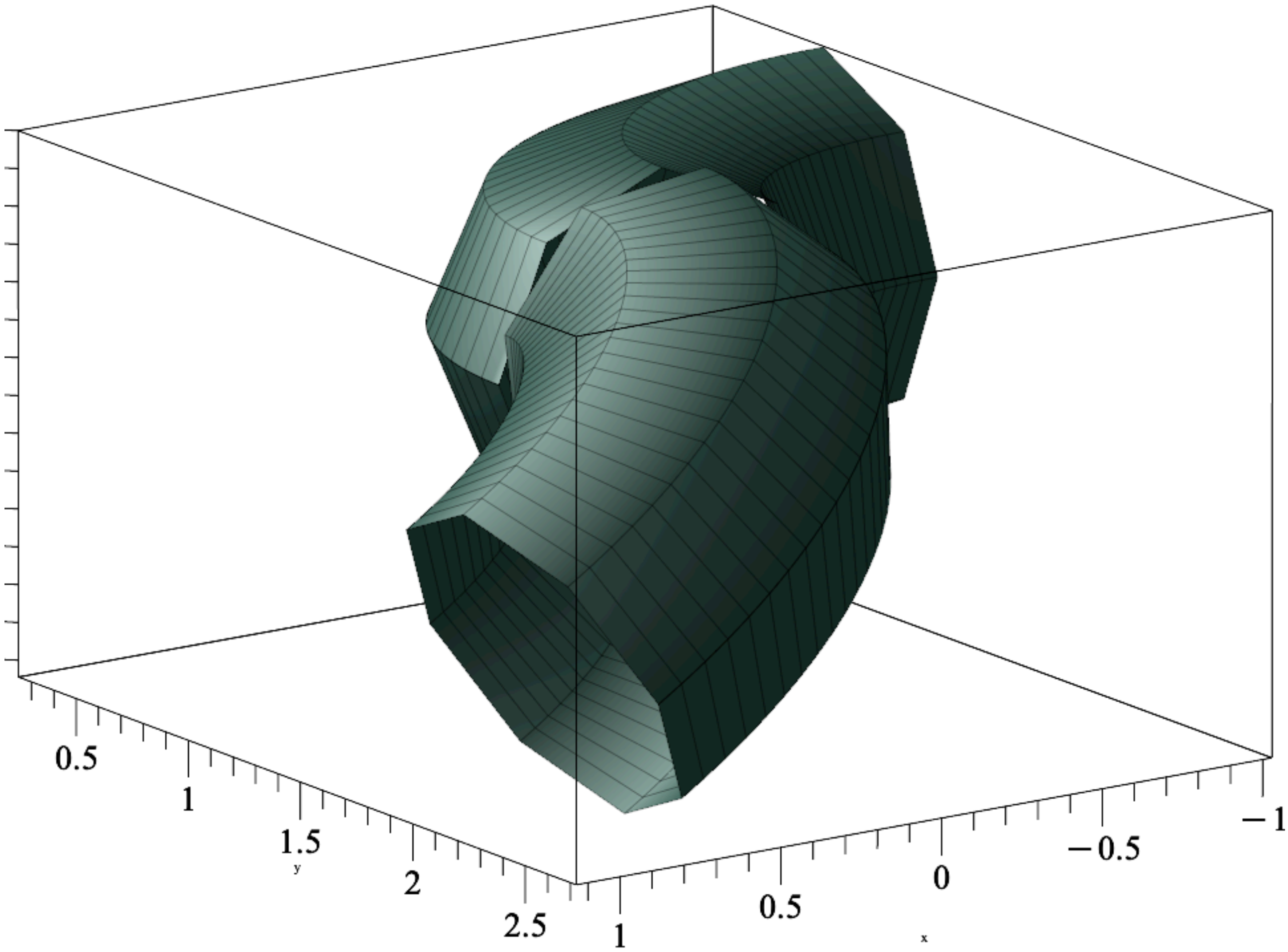
$$\theta = \frac{\pi}{3} \qquad (2.1.2)$$

Det lodrette stykke roteres ved at multiplicere vektorfunktionen for cirkelstykkerne med rotationsmatricen R_y ved hjælp af prikproduktet. Dette udføres komponentvist, således at hver række i R_y anvendes til at danne et nyt vektorelement. Denne fremgangsmåde gør det muligt at multiplicere hvert vektorkoordinat med en konstant.

```
> cirk2rot:= t-> [Ry(theta)[1].Vector(cirk2(t))*d,Ry(theta)[2].Vector(cirk2(t))*c,Ry(theta)[3].Vector(cirk2(t))*d]:'cirk2rot(t)'=cirk2rot(t)
```

$$cirk2rot(t) = \left[-0.511500\sqrt{3}\,(\sin(t) - 1), 1.484 + 0.742\cos(t), 0.511500\sin(t) - 0.511500\right] \qquad (2.1.3)$$

```
> cirk2rotFig:= display(labels=["x","y","z"], tubeplot(cirk2rot(t),radius=rad, t=0....Pi/2,  color=mg, scaling=constrained)):
> display(cirk1Fig, cirk2rotFig);
```

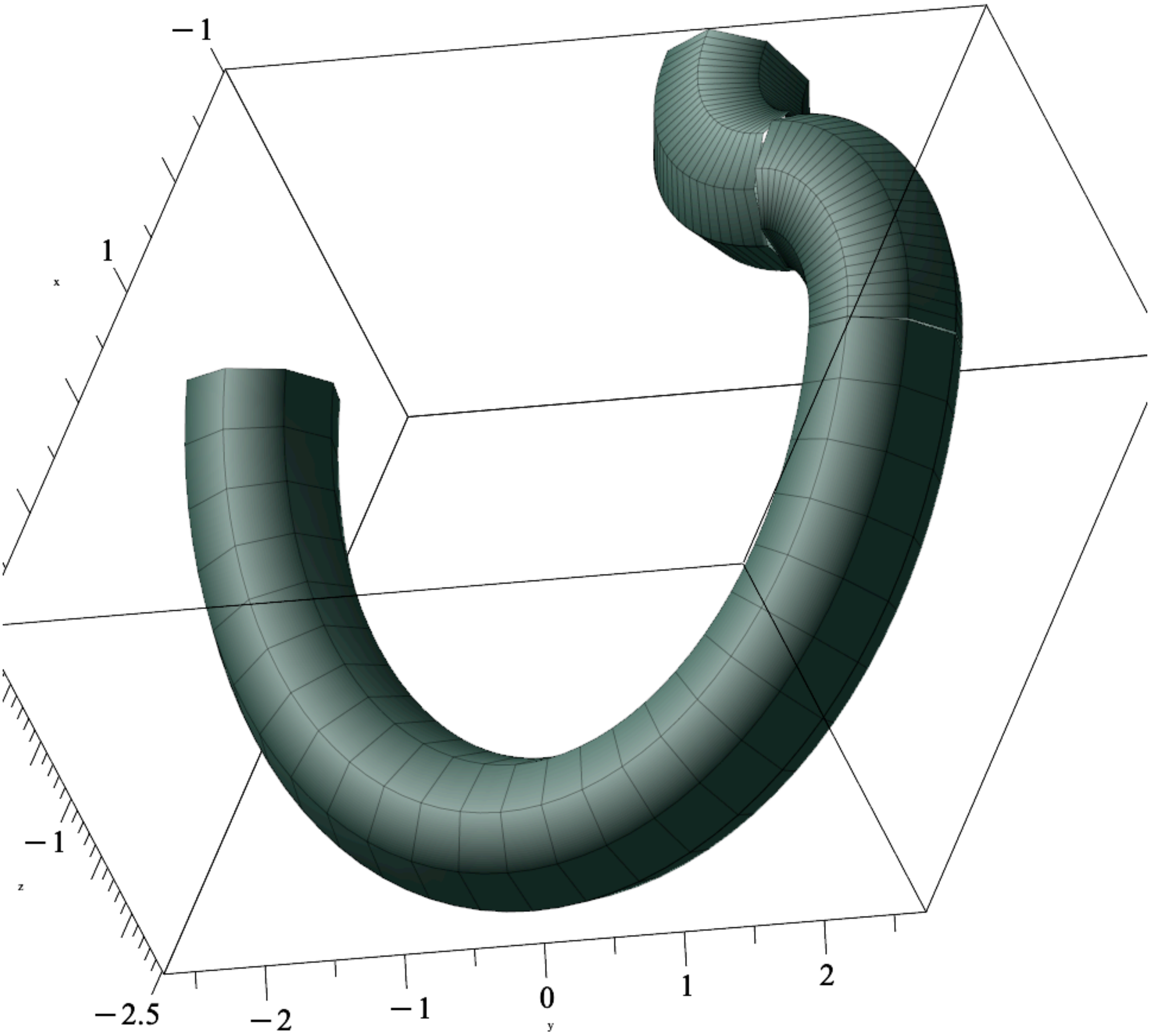



```
> cirk3rot:= t-> [Ry(theta)[1].Vector(cirk3(t))*d,Ry(theta)[2].
  Vector(cirk3(t))*c,Ry(theta)[3].Vector(cirk3(t))*d]:'cirk3rot(t)
```

```
'=cirk3rot(t);
```

$$cirk3rot(t) = \left[-0.511500\sqrt{3}\,(-3\sin(t)-1), 2.226\cos(t), -1.53450\sin(t) - 0.511500\right] \quad \textbf{(2.1.4)}$$

```
> cirk3rotFig:= tubeplot(cirk3rot(t),radius=rad, t=0...Pi,  color=
  mg, scaling=constrained):
> display(cirk1Fig, cirk2rotFig, cirk3rotFig,orientation=[-6,46,
  -33]);
```

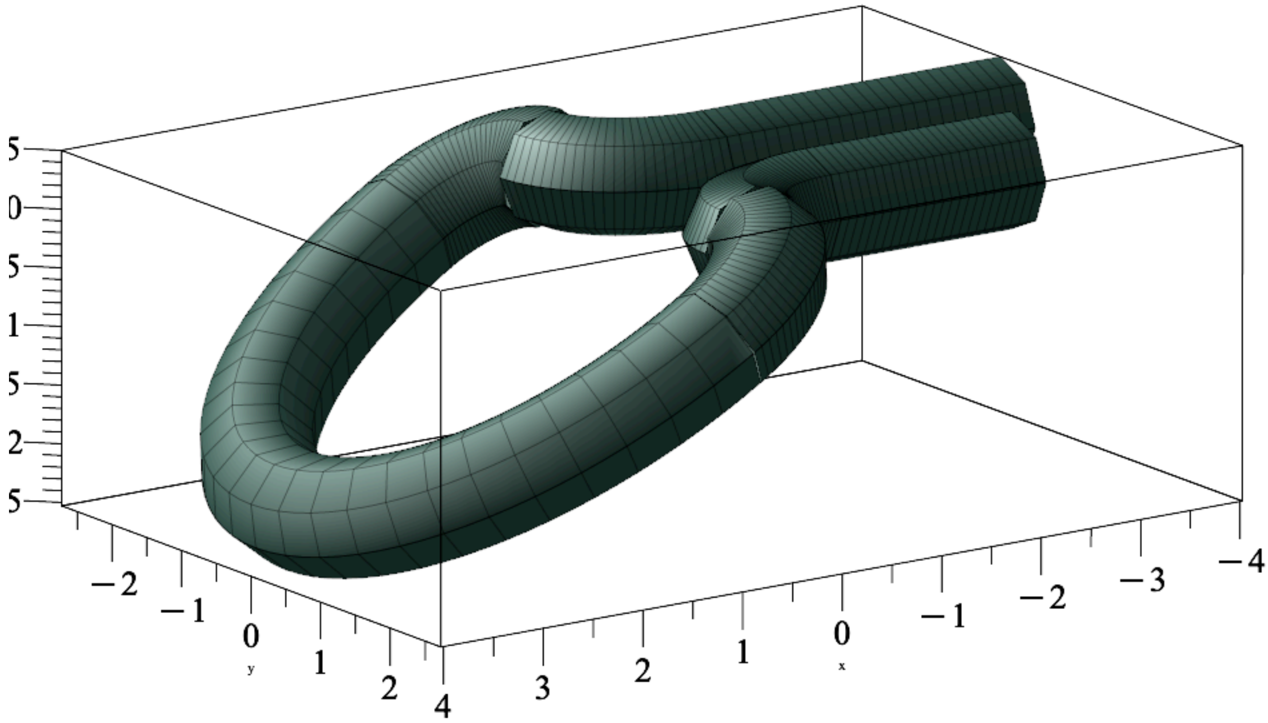



```
> cirk4rot:= t-> [Ry(theta)[1].Vector(cirk4(t))*d,Ry(theta)[2].
  Vector(cirk4(t))*c,Ry(theta)[3].Vector(cirk4(t))*d]:'cirk4rot(t)
```

```
'=cirk4rot(t)
```

$$cirk4rot(t) = \left[-0.511500\sqrt{3}\,(\sin(t)-1), -1.484-0.742\cos(t), 0.511500\sin(t) - 0.511500\right] \quad \textbf{(2.1.5)}$$

```
> cirk4rotFig:= tubeplot(cirk4rot(t),radius=rad, t=0....Pi/2,
  color=mg, scaling=constrained):
> display(cirk1Fig, cirk2rotFig, cirk3rotFig, cirk4rotFig,
  cirk5Fig, cyl1, cyl2);
```



C.3 Den anden løkke

Der konstrueres vektorfunktioner for fire kvarte cirkler, én halvcirkel samt to vandrette, lineære funktioner.

```
> cirk6:= t-> [(1-cos(t)+e)*d, (2-sin(t)+f)*c,g*d]:'cirk6(t)
  '=cirk6(t)
```

$$cirk6(t) = [0.867504 - 1.023 \cos(t), 1.484 - 0.742 \sin(t), -1.01891] \quad \textbf{(3.1)}$$

```
> cirk6Fig:= display( tubeplot(cirk6(t),radius=rad, t=0....Pi/2,
  color=lg, scaling=constrained)):
```

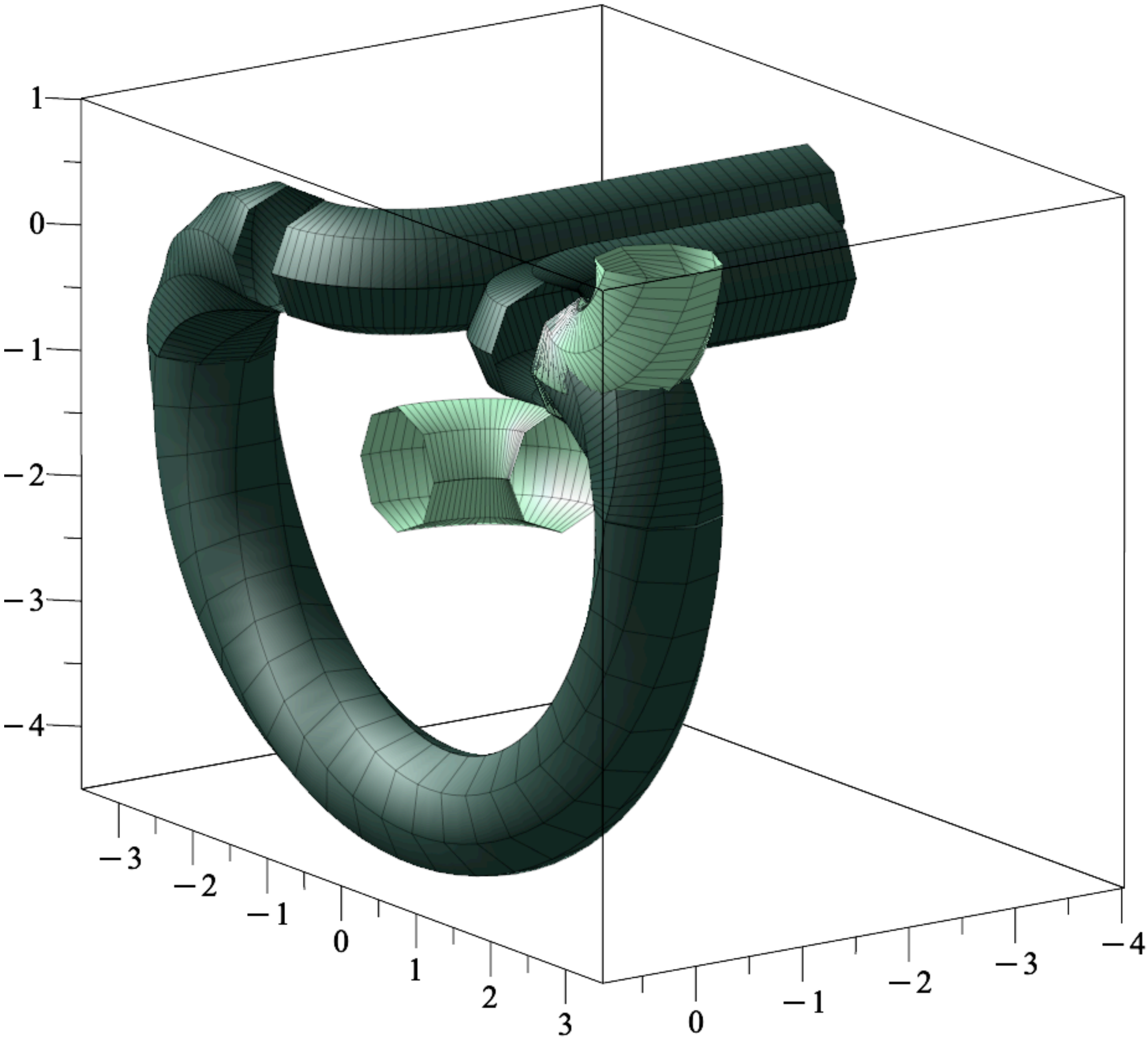
```
> cirk7:= t-> [0, (2+cos(t)), -sin(t)+1]:'cirk7(t)'=cirk7(t)
```

$$cirk7(t) = [0, 2 + \cos(t), -\sin(t) + 1] \quad \textbf{(3.2)}$$

```
> cirk7Fig:= display( tubeplot(cirk7(t),radius=rad, t=0....Pi/2,
  color=lg, scaling=constrained)):
```

Kurvedelene samles endnu ikke, da der senere tilføjes konstanter, som skal sikre korrekt sammenhæng mellem delene.

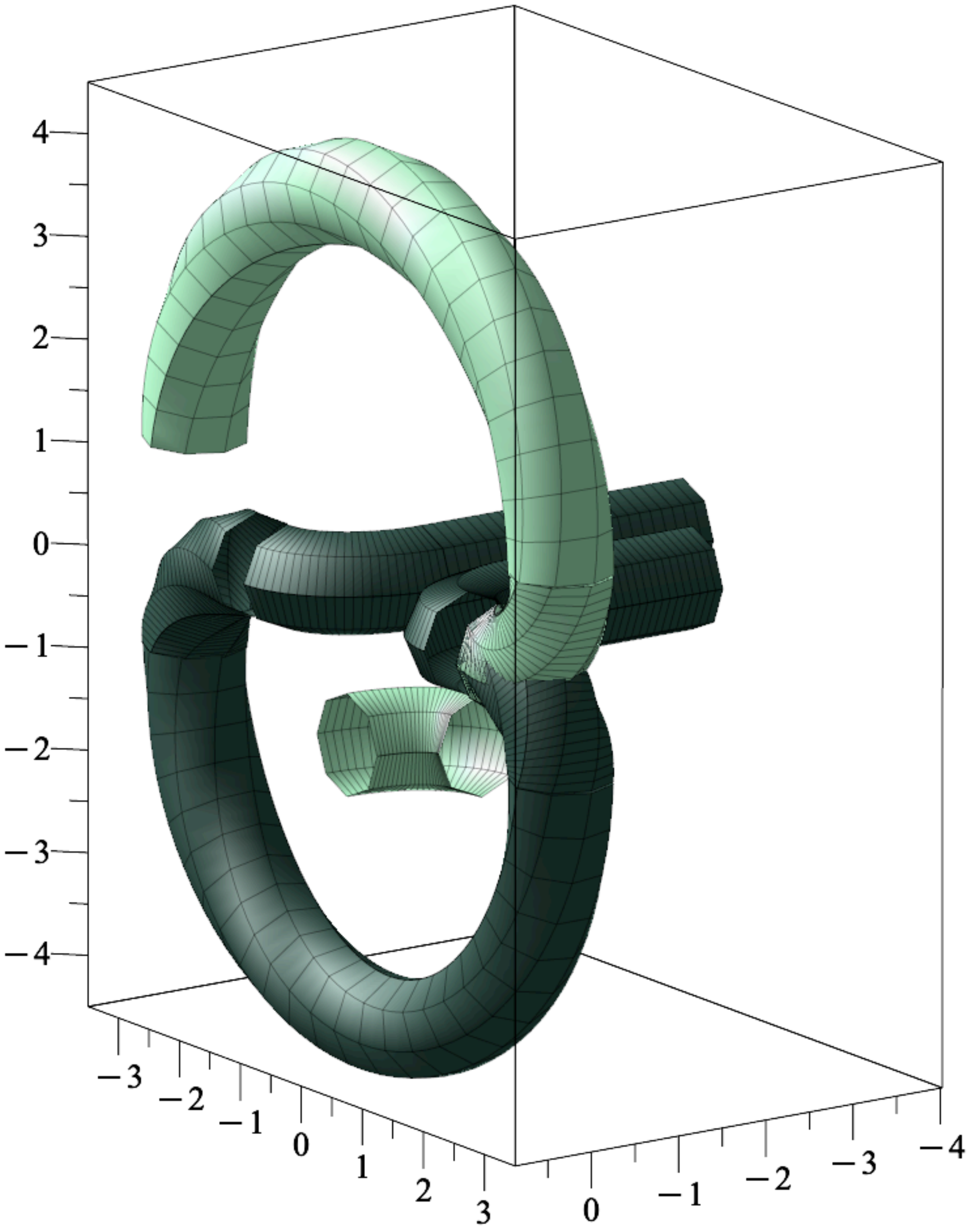
```
> display(display(cirk1Fig, cirk2Fig, cirk3Fig, cirk4Fig,
  cirk5Fig, cyl1, cyl2,cirk6Fig, cirk7Fig));
```




```
> cirk8:= t-> [0, -3*cos(t), 1+3*sin(t)]:'cirk8(t)'=cirk8(t)
```

$$cirk8(t) = [0, -3\cos(t), 1 + 3\sin(t)] \tag{3.3}$$

```
> cirk8Fig:= display( tubeplot(cirk8(t),radius=rad, t=0...Pi,
  color=lg, scaling=constrained)):
> display(display(cirk1Fig, cirk2Fig, cirk3Fig, cirk4Fig,
  cirk5Fig, cyl1, cyl2,cirk6Fig, cirk7Fig,cirk8Fig));
```

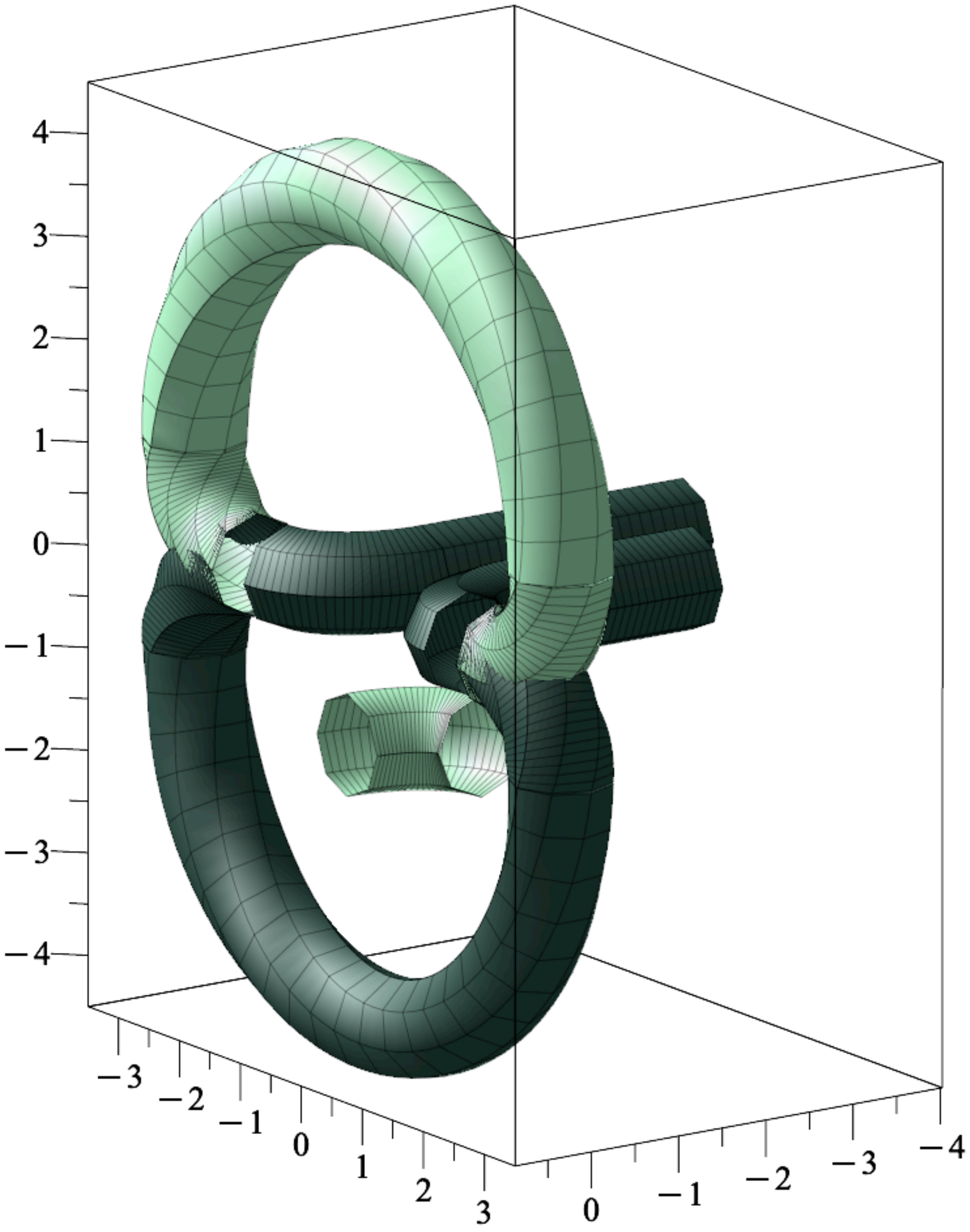



```
> cirk9:= t->  [0, -2-cos(t), 1-sin(t)];:'cirk8(t)'=cirk8(t)
```

$$cirk9 := t \mapsto [0, -2 + (-\cos(t)), 1 + (-\sin(t))]$$

$$cirk8(t) = [0, -3\cos(t), 1 + 3\sin(t)] \quad \textbf{(3.4)}$$

```
> cirk9Fig:= display( tubeplot(cirk9(t),radius=rad, t=0....Pi/2,
  color=lg, scaling=constrained)):
> display(display(cirk1Fig, cirk2Fig, cirk3Fig, cirk4Fig,
  cirk5Fig, cyl1, cyl2,cirk6Fig, cirk7Fig,cirk8Fig,cirk9Fig));
```

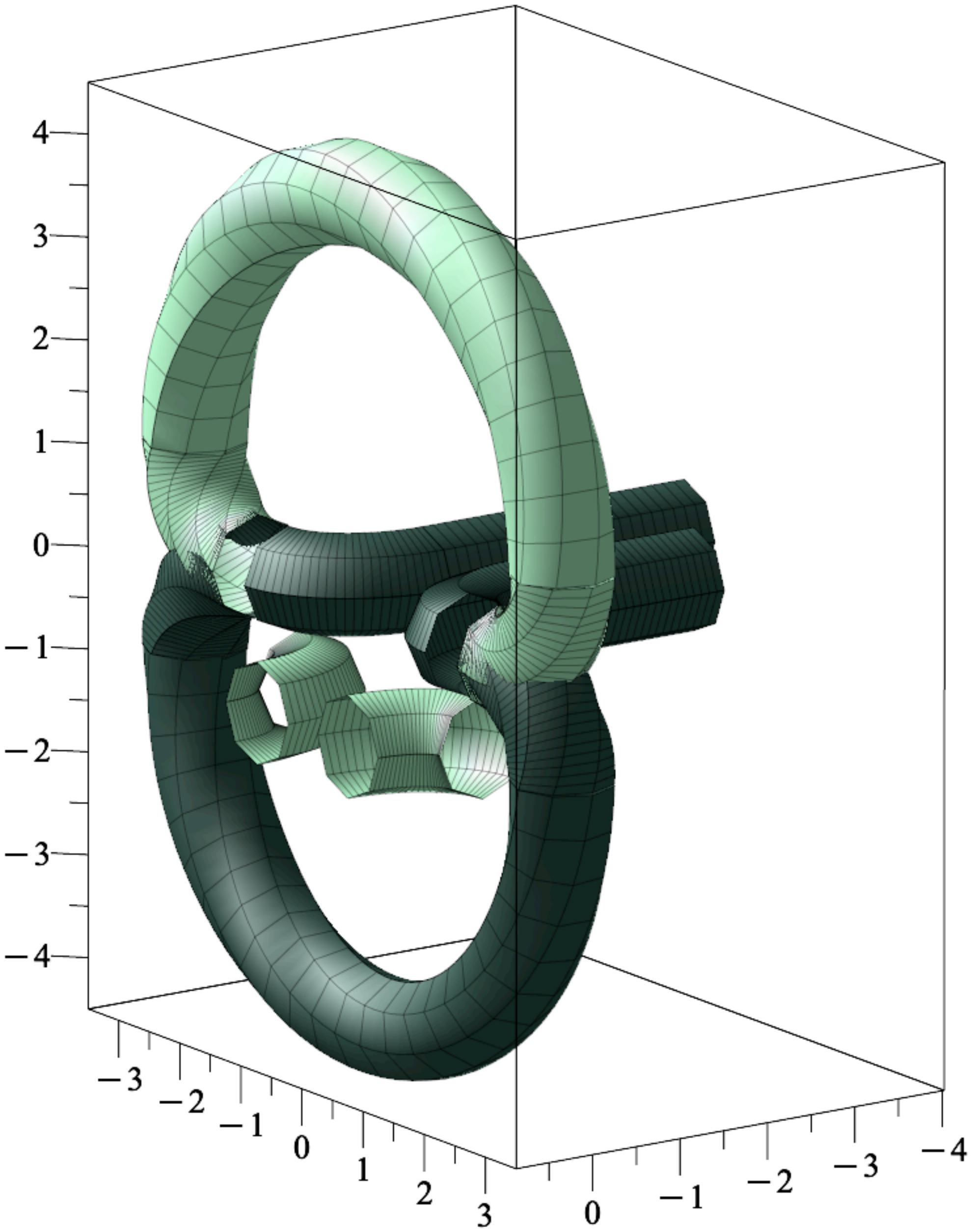



```
> cirk10:= t-> [(e+1-cos(t))*d, (-2+sin(t)+f)*c, g*d];:'cirk10(t)
  '=cirk10(t)
```

$$cirk10 := t \mapsto [(e + 1 + (-\cos(t)))\cdot d, (-2 + \sin(t) + f)\cdot c, g\cdot d]$$

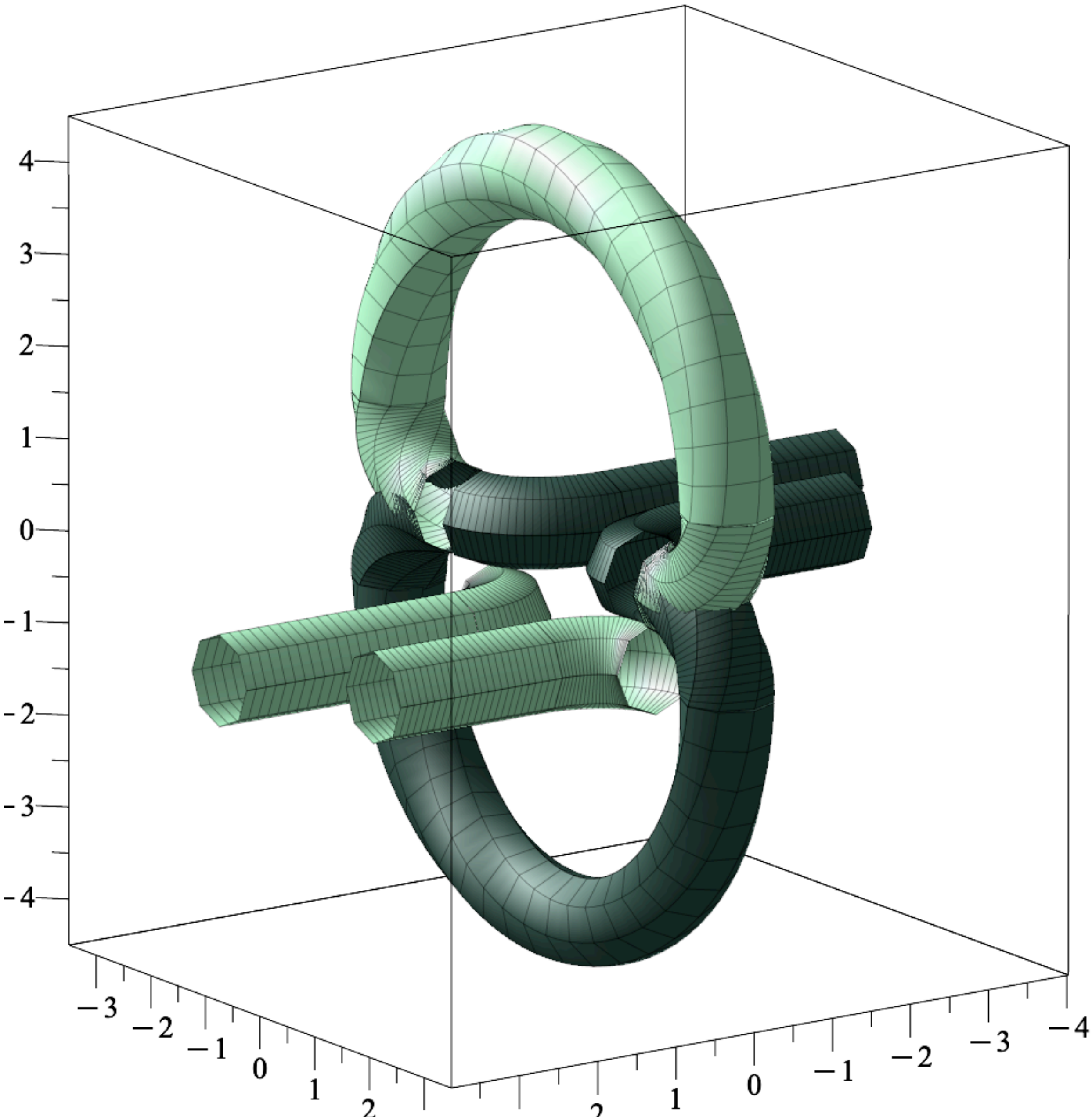
$$cirk10(t) = [0.867504 - 1.023\cos(t), -1.484 + 0.742\sin(t), -1.01891] \tag{3.5}$$

```
> cirk10Fig:= display( tubeplot(cirk10(t),radius=rad, t=0....Pi/2,
   color=lg, scaling=constrained)):
> display(display(cirk1Fig, cirk2Fig, cirk3Fig, cirk4Fig,
  cirk5Fig, cyl1, cyl2,cirk6Fig, cirk7Fig,cirk8Fig,cirk9Fig,
  cirk10Fig));
```




```
> cyl3:= tubeplot([t+d+e*d, c+f*c,g*d], radius=rad, t=0...i,
  color=lg, scaling=constrained):
```

```
> cyl4:= tubeplot([t+d+e*d, -c+f*c,(0*d)+g*d], radius=rad, t=0...
  j, color=lg, scaling=constrained):
> display(display(cirk1Fig, cirk2Fig, cirk3Fig, cirk4Fig,
  cirk5Fig, cyl1, cyl2,cirk6Fig, cirk7Fig,cirk8Fig,cirk9Fig,
  cirk10Fig,cyl3, cyl4));
```



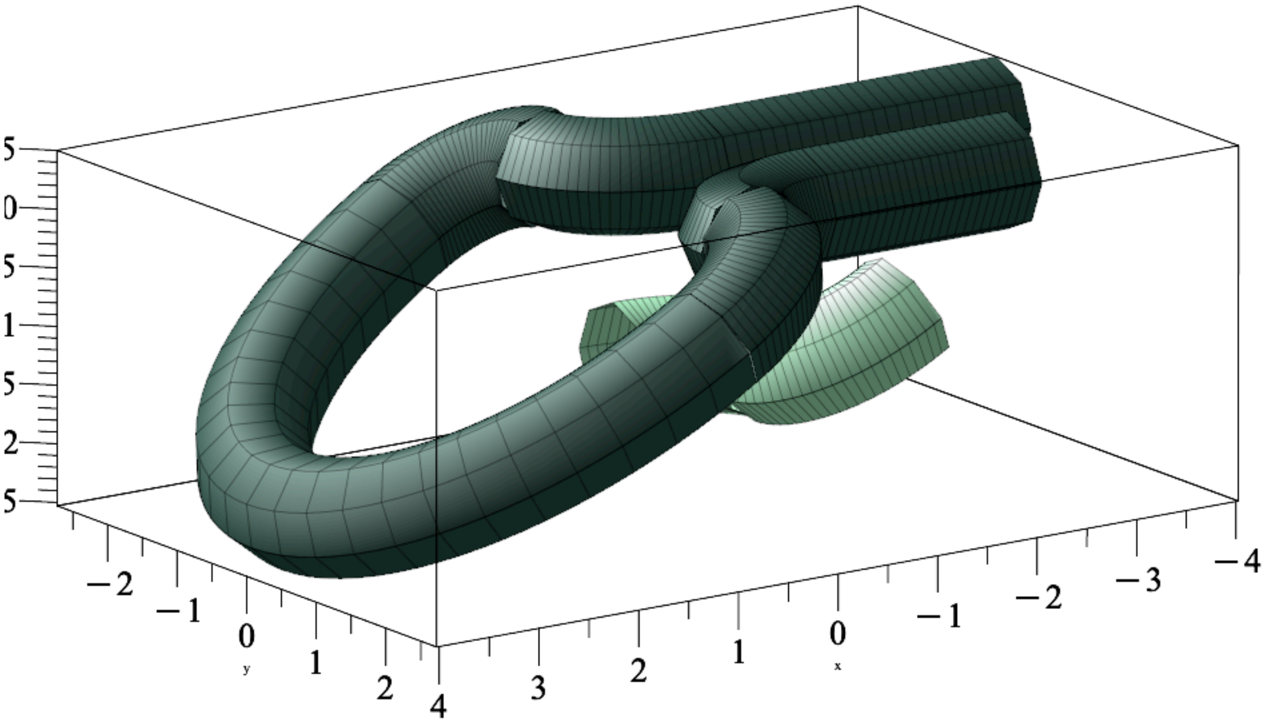
C.3.1 Rotation af lodret del af løkken

Det lodrette stykke roteres ved at multiplicere vektorfunktionen for cirkelstykkerne med rotationsmatricen R_y ved hjælp af prikproduktet. Dette udføres komponentvist, således at hver række i R_y anvendes til at danne et nyt vektorelement. Denne fremgangsmåde gør det muligt at multiplicere +g addere hvert vektorkoordinat med en konstant.

```
> cirk7rot:= t-> [(Ry(theta)[1].Vector(cirk7(t))+e)*d,(Ry(theta)
  [2].Vector(cirk7(t))+f)*c,(Ry(theta)[3].Vector(cirk7(t))+g)*d]
  :'cirk7rot(t)'=cirk7rot(t)
```

$$cirk7rot(t) = \left[-0.511500\sqrt{3}\,(-\sin(t)+1) - 0.155496,\ 1.484 + 0.742\cos(t),\ -0.511500\sin(t) - 0.507408\right] \tag{3.1.1}$$

```
> cirk7rotFig:= display( tubeplot(cirk7rot(t),radius=rad, t=0....
  Pi/2,  color=lg, scaling=constrained)):
> display(cirk1Fig, cirk2rotFig, cirk3rotFig, cirk4rotFig,
  cirk5Fig, cyl1, cyl2,cirk6Fig, cirk7rotFig);
```

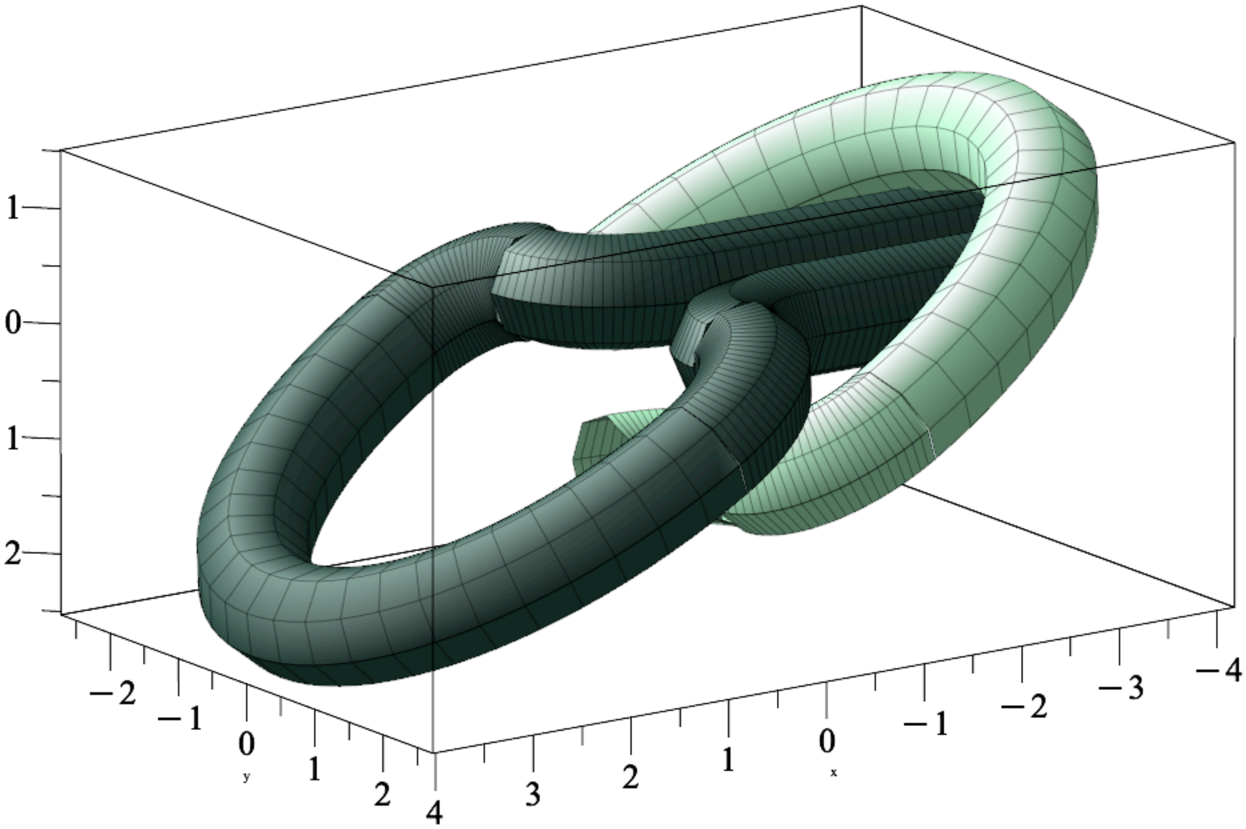



```
> cirk8rot:= t-> [(Ry(theta)[1].Vector(cirk8(t))+e)*d,(Ry(theta)
  [2].Vector(cirk8(t))+f)*c,(Ry(theta)[3].Vector(cirk8(t))+g)*d]
```

```
:'cirk8rot(t)'=cirk8rot(t)
```

$$cirk8rot(t) = \left[-0.511500\sqrt{3}\,(1 + 3\sin(t)) - 0.155496, -2.226\cos(t), -0.507408 + 1.53450\sin(t)\right] \tag{3.1.2}$$

```
> cirk8rotFig:= display( tubeplot(cirk8rot(t),radius=rad, t=0...
  Pi,  color=lg, scaling=constrained)):
> display(cirk1Fig, cirk2rotFig, cirk3rotFig, cirk4rotFig,
  cirk5Fig, cyl1, cyl2,cirk6Fig, cirk7rotFig,cirk8rotFig);
```

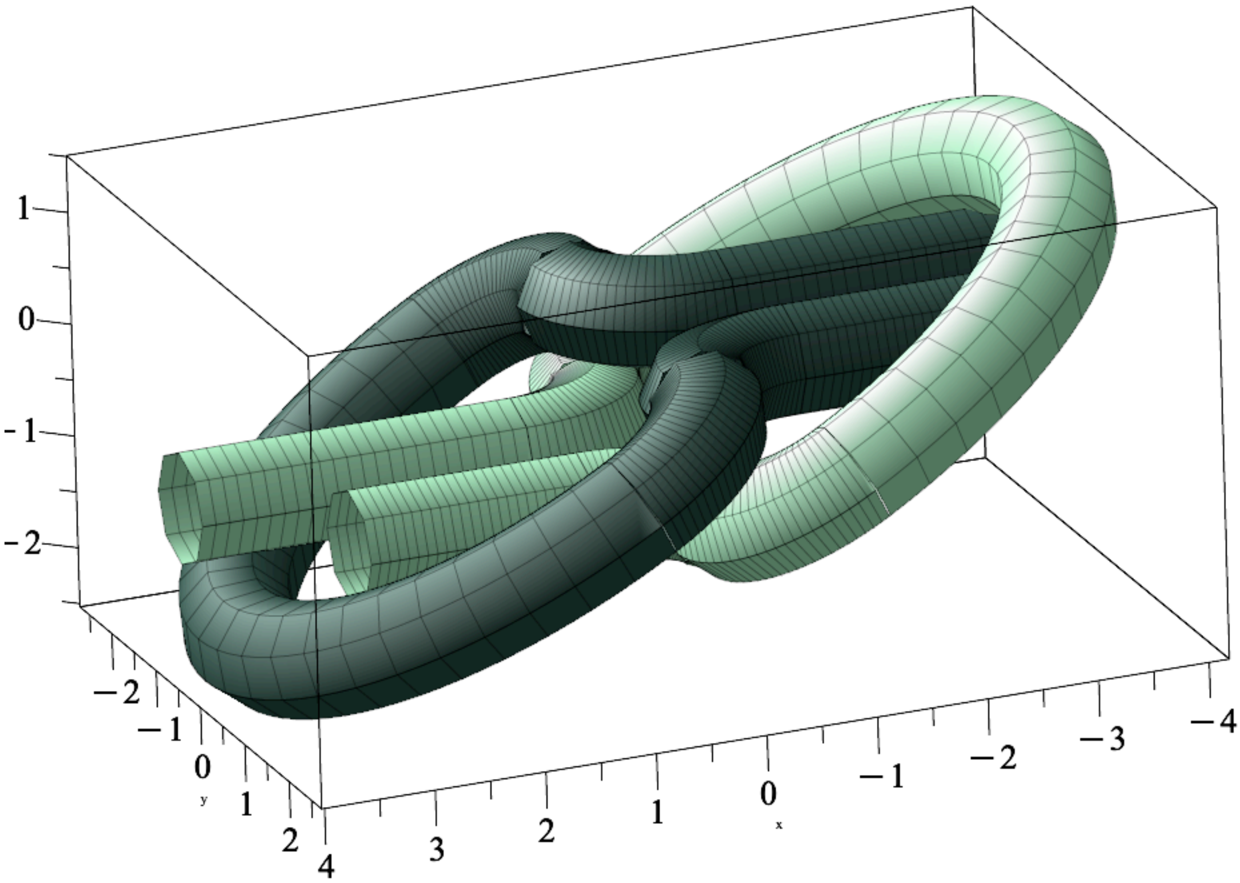



```
> cirk9rot:= u-> [(Ry(theta)[1].Vector(cirk9(t))+e)*d,(Ry(theta)
  [2].Vector(cirk9(t))+f)*c,(Ry(theta)[3].Vector(cirk9(t))+g)*d]
```

```
:'cirk9rot(t)'=cirk9rot(t)
```

$$cirk9rot(t) = \left[-0.511500\sqrt{3}\,(-\sin(t)+1) - 0.155496, -1.484 - 0.742\cos(t), -0.511500\sin(t) - 0.507408\right] \quad \textbf{(3.1.3)}$$

```
> cirk9rotFig:= display(tubeplot(cirk9rot(t),radius=rad, t=0....
  Pi/2,  color=lg, scaling=constrained)):
> display(cirk1Fig, cirk2rotFig, cirk3rotFig, cirk4rotFig,
  cirk5Fig, cyl1, cyl2,cirk6Fig, cirk7rotFig, cirk8rotFig,
  cirk9rotFig, cirk10Fig, cyl3, cyl4);
```



C.4 rektangulært objet sweepet om kurven

C.4.1 Objektet

Et rektangulært objekt bestemme, men centum i fodpunktskurven. Der bestemme et objekt for hver flade.

```
> w1:= u -> <0,-a/2,u/2>:'w1(t)'=w1(u);
```

$$w1(t) = \begin{bmatrix} 0 \\ -\frac{1}{2} \\ \frac{u}{2} \end{bmatrix} \tag{4.1.1}$$

```
> w2:= u -> <0,a/2,u/2>:'w2(t)'=w2(u);
```

$$w2(t) = \begin{bmatrix} 0 \\ \frac{1}{2} \\ \frac{u}{2} \end{bmatrix} \tag{4.1.2}$$

```
> w3:= u -> <0,u/2,b/2>:'w3(t)'=w3(u);
```

$$w3(t) = \begin{bmatrix} 0 \\ \frac{u}{2} \\ \frac{1}{2} \end{bmatrix} \tag{4.1.3}$$

```
> w4:= u -> <0,u/2,-b/2>:'w4(t)'=w4(u);
```

$$w4(t) = \begin{bmatrix} 0 \\ \frac{u}{2} \\ -\frac{1}{2} \end{bmatrix} \tag{4.1.4}$$

```
> display(plot3d(w1(u),u=-a...a,scaling = constrained,color=lg),
  plot3d(w2(u),u=-a...a,scaling = constrained,color=mg),plot3d(w3
  (u),u=-b...b,scaling = constrained,color=mg),plot3d(w4(u),u=-b..
  .b,scaling = constrained,color=lg,labels=["x","y","z"])):
```

C.4.2 Cirk1(t)

Der bestemmes en rotationsmatrix, som angiver, hvordan objektet skal roteres langs fodpunktskurven. Rotationsmatricen udgøres her af Frenet–Serret-baserne.

```
> R1:= t -> <FSt(cirk1,[t])[1],FSn(cirk1,[t])[1],FSb(cirk1,[t])[1]
  ;FSt(cirk1,[t])[2],FSn(cirk1,[t])[2],FSb(cirk1,[t])[2];FSt
  (cirk1,[t])[3],FSn(cirk1,[t])[3],FSb(cirk1,[t])[3]>: 'R1(t)'=R1
  (t)
```

$$R1(t) = \begin{bmatrix} -\frac{1.023\sin(t)}{\sqrt{0.798547 - 0.247983\cos(2t)}} & -\frac{0.74200}{\sqrt{0.798547 - 0.}} \cdots \\ -\frac{0.742\cos(t)}{\sqrt{0.798547 - 0.247983\cos(2t)}} & \frac{1.02300}{\sqrt{0.798547 - 0.2}} \cdots \\ 0 & 0. \cdots \end{bmatrix} \quad \mathbf{(4.2.1)}$$

Til konstruktionen anvendes formlen for Cosserat sweeping, som gør det muligt at generere en rumlig kurve, hvor objektet roteres kontinuerligt omkring fodpunktskurven. For hver flade af objektet dannes en separat kurve ved at anvende samme sweeping-princip.

```
> q11:= (t,u) -> convert((R1(t).w1(u))^(%T),list)+cirk1(t): 'q11
  (t,u)'=q11(t,u)
```

$$q11(t,u) = \left[-1.023 + 1.023\cos(t) + \frac{0.371000\cos(t)}{\sqrt{0.798547 - 0.247983\cos(2t)}}, 1.484 - 0.742\sin(t) - \frac{0.511500\sin(t)}{\sqrt{0.798547 - 0.247983\cos(2t)}}, -0.500000\,u\right] \quad \mathbf{(4.2.2)}$$

```
> qp11:=plot3d(q11(t,u),t=0..Pi/2, u=-a...a,color=lg):
> q12:= (t,u) -> convert((R1(t).w2(u))^(%T),list)+cirk1(t): 'q12
  (t,u)'=q12(t,u)
```

$$q12(t,u) = \left[-1.023 + 1.023\cos(t) - \frac{0.371000\cos(t)}{\sqrt{0.798547 - 0.247983\cos(2t)}}, 1.484 - 0.742\sin(t) + \frac{0.511500\sin(t)}{\sqrt{0.798547 - 0.247983\cos(2t)}}, -0.500000\,u\right] \quad \mathbf{(4.2.3)}$$

```
> qp12:=plot3d(q12(t,u),t=0..Pi/2, u=-a...a,color=lg):
> q13:= (t,u) -> convert((R1(t).w3(u))^(%T),list)+cirk1(t): 'q13
  (t,u)'=q13(t,u)
```

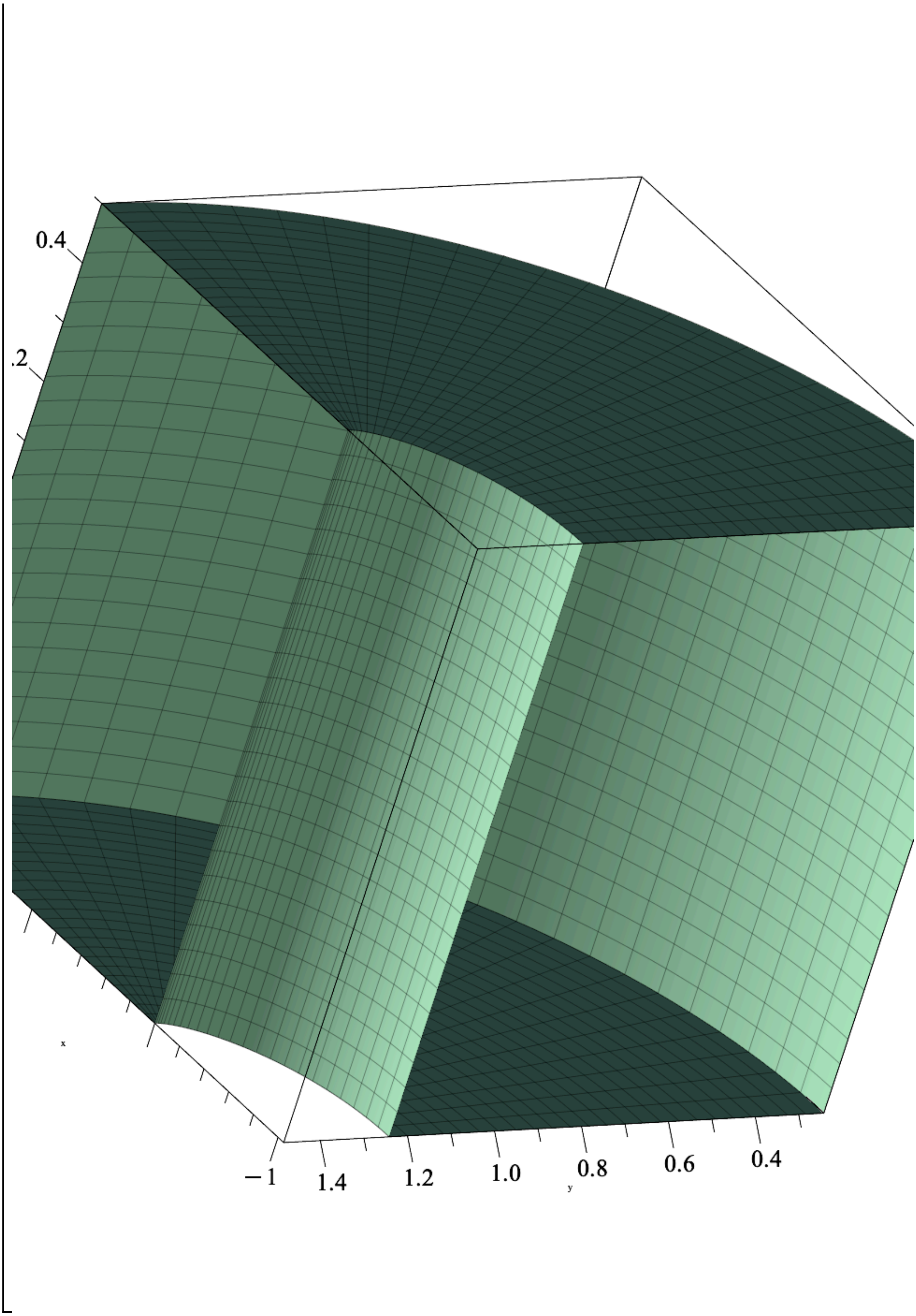
$$q13(t,u) = \left[-1.023 + 1.023\cos(t) - \frac{0.371000\cos(t)\,u}{\sqrt{0.798547 - 0.247983\cos(2t)}}, 1.484 - 0.742\sin(t) + \frac{0.511500\sin(t)\,u}{\sqrt{0.798547 - 0.247983\cos(2t)}}, -0.500000\right] \quad \mathbf{(4.2.4)}$$

```
> qp13:=plot3d(q13(t,u),t=0..Pi/2, u=-b...b,color=mg):
> q14:= (t,u) -> convert((R1(t).w4(u))^(%T),list)+cirk1(t): 'q14
  (t,u)'=q14(t,u)
```

$$q14(t,u) = \left[-1.023 + 1.023\cos(t) - \frac{0.371000\cos(t)\,u}{\sqrt{0.798547 - 0.247983\cos(2t)}}, 1.484\right. \quad \mathbf{(4.2.5)}$$

$$-0.742\sin(t)+\frac{0.511500\sin(t)\,u}{\sqrt{0.798547-0.247983\cos(2\,t)}},\,0.500000\Big]$$

```
> qp14:=plot3d(q14(t,u),t=0..Pi/2, u=-b...b,color=mg):
> display(qp11,qp12,qp13,qp14,labels=["x","y","z"])
```


0.4
.2
x
−1
1.4
1.2
1.0
0.8
y
0.6
0.4

C.4.3 Cirk2rot(t)

Der bestemmes en rotationsmatrix, som angiver, hvordan objektet skal roteres langs fodpunktskurven. Rotationsmatricen bestemmes således, at objektets endeposition stemmer overens med kurvernes endepositioner både før og efter rotationen.

```
> R2:= t -> <0.500000*sqrt(3)*cos(t), sin(t), -0.500000*cos(t);-
  sin(t), -cos(t), 0.;0, 0.,-0.133972*sin(t)-0.866028>:'R2(t)'=R2
  (t):
```

Til konstruktionen anvendes formlen for Cosserat sweeping, som gør det muligt at generere en rumlig kurve, hvor objektet roteres kontinuerligt omkring fodpunktskurven. For hver flade af objektet dannes en separat kurve ved at anvende samme sweeping-princip.

```
> q21:= (t,u) -> convert((R2(t).w1(u))^(%T),list)+cirk2rot(t):
  'q21(t,u)'=q21(t,u)
```

$$q21(t,u) = \left[-0.511500\sqrt{3}\,(\sin(t)-1) - \frac{\sin(t)}{2} - 0.250000\cos(t)\,u,\ 1.484 + 1.24200\cos(t),\ 0.511500\sin(t) - 0.511500 + \frac{(-0.133972\sin(t)-0.866028)\,u}{2}\right] \quad (4.3.1)$$

```
> qp21:=plot3d(q21(t,u),t=0..Pi/2, u=-a...a,color=lg):
> q22:= (t,u) -> convert((R2(t).w2(u))^(%T),list)+cirk2rot(t):
  'q22(t,u)'=q22(t,u)
```

$$q22(t,u) = \left[-0.511500\sqrt{3}\,(\sin(t)-1) + \frac{\sin(t)}{2} - 0.250000\cos(t)\,u,\ 1.484 + 0.242000\cos(t),\ 0.511500\sin(t) - 0.511500 + \frac{(-0.133972\sin(t)-0.866028)\,u}{2}\right] \quad (4.3.2)$$

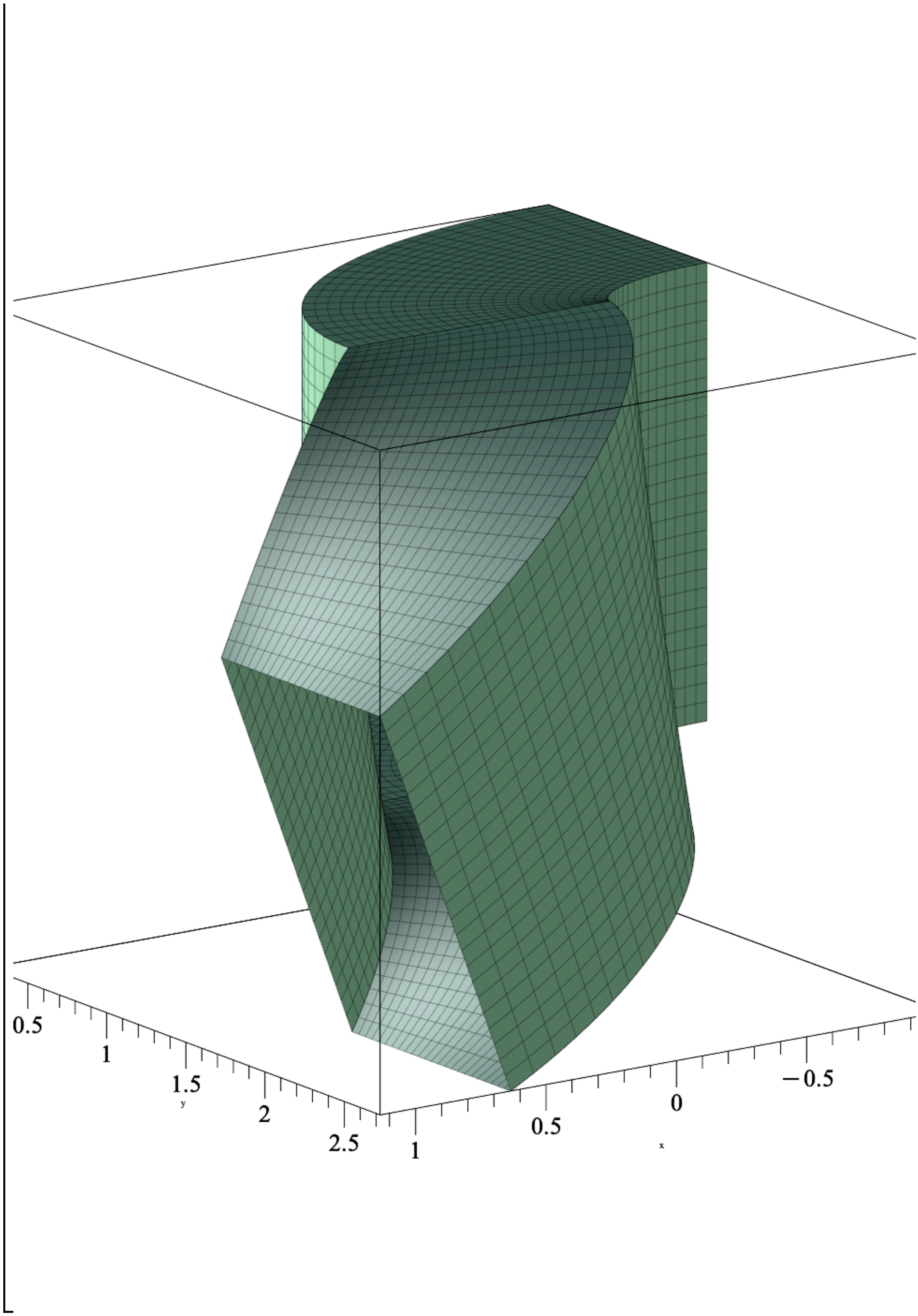
```
> qp22:=plot3d(q22(t,u),t=0..Pi/2, u=-a...a,color=lg):
> q23:= (t,u) -> convert((R2(t).w3(u))^(%T),list)+cirk2rot(t):
  'q23(t,u)'=q23(t,u)
```

$$q23(t,u) = \left[-0.511500\sqrt{3}\,(\sin(t)-1) + \frac{\sin(t)\,u}{2} - 0.250000\cos(t),\ 1.484 + 0.742\cos(t) - \frac{\cos(t)\,u}{2},\ 0.444514\sin(t) - 0.944514\right] \quad (4.3.3)$$

```
> qp23:=plot3d(q23(t,u),t=0..Pi/2, u=-b...b,color=mg):
> q24:= (t,u) -> convert((R2(t).w4(u))^(%T),list)+cirk2rot(t):
  'q24(t,u)'=q24(t,u)
```

$$q24(t,u) = \left[-0.511500\sqrt{3}\,(\sin(t)-1) + \frac{\sin(t)\,u}{2} + 0.250000\cos(t),\ 1.484 + 0.742\cos(t) - \frac{\cos(t)\,u}{2},\ 0.578486\sin(t) - 0.078486\right] \quad (4.3.4)$$

```
> qp24:=plot3d(q24(t,u),t=0..Pi/2, u=-b...b,color=mg):
> display(qp11,qp12,qp13,qp14,qp21,qp22,qp23,qp24, labels=["x",
  "y","z"])
```



C.4.4 Cirk3rot(t)

Der bestemmes en rotationsmatrix, som angiver, hvordan objektet skal roteres langs fodpunktskurven. Rotationsmatricen bestemmes således, at objektets rotation i endepositionerne er identisk, og at den resulterende kurve så vidt muligt svarer til kurveforløbet i Grasshopper.

```
> R3:= t -> <-(cos(t))^2*0.500000*sqrt(3),-0.999999*sin(t), -(cos
  (t))^2*0.5;-sin(t)*0.866028, -cos(t), 0.500000*sin(t);-0.500001,
  0.,-0.866028>:'R3(t)'=R3(t)
```

En rotationsmatricen for cirk3rot(t) der udgøres af Frenet–Serret-baserne.

```
> FSB:= t -> <FSt(cirk3rot,[t])[1],FSn(cirk3rot,[t])[1],FSb
  (cirk3rot,[t])[1];FSt(cirk3rot,[t])[2],FSn(cirk3rot,[t])[2],FSb
  (cirk3rot,[t])[2];FSt(cirk3rot,[t])[3],FSn(cirk3rot,[t])[3],FSb
  (cirk3rot,[t])[3]>: 'FSB(t)'=FSB(t)
```

$$FSB(t) = \quad \textbf{(4.4.1)}$$

$$\begin{bmatrix} \frac{1.53450\sqrt{3}\cos(t)}{\sqrt{7.18693+2.23184\cos(2t)}} & -\frac{1.92778 \sin}{\sqrt{7.18693+2.2318}} \cdots \\ -\frac{2.226\sin(t)}{\sqrt{7.18693+2.23184\cos(2t)}} & -\frac{3.06901 \cos}{\sqrt{7.18693+2.2318}} \cdots \\ -\frac{1.53450\cos(t)}{\sqrt{7.18693+2.23184\cos(2t)}} & \frac{1.11300 \sin(}{\sqrt{7.18693+2.2318}} \cdots \end{bmatrix}$$

Til konstruktionen anvendes formlen for Cosserat sweeping, som gør det muligt at generere en rumlig kurve, hvor objektet roteres kontinuerligt omkring fodpunktskurven. For hver flade af objektet dannes en separat kurve ved at anvende samme sweeping-princip.

```
> q31:= (t,u) -> convert((R3(t).w1(u))^(%T),list)+cirk3rot(t):
  'q31(t,u)'=q31(t,u)
```

$$q31(t,u) = \left[-0.511500\sqrt{3}\,(-3\sin(t)-1)+0.500000\sin(t)-0.250000\cos(t)^2 u,\; 2.72600\cos(t)+0.250000\sin(t)\,u,\; -1.53450\sin(t)-0.511500-0.433014\,u\right] \quad \textbf{(4.4.2)}$$

```
> qp31:=plot3d(q31(t,u),t=0..Pi, u=-a...a,color=lg):
> q32:= (t,u) -> convert((R3(t).w2(u))^(%T),list)+cirk3rot(t):
  'q32(t,u)'=q32(t,u)
```

$$q32(t,u) = \left[-0.511500\sqrt{3}\,(-3\sin(t)-1)-0.500000\sin(t)-0.250000\cos(t)^2 u,\; 1.72600\cos(t)+0.250000\sin(t)\,u,\; -1.53450\sin(t)-0.511500-0.433014\,u\right] \quad \textbf{(4.4.3)}$$

```
> qp32:=plot3d(q32(t,u),t=0..Pi, u=-a...a,color=lg):
> q33:= (t,u) -> convert((R3(t).w3(u))^(%T),list)+cirk3rot(t):
  'q33(t,u)'=q33(t,u)
```

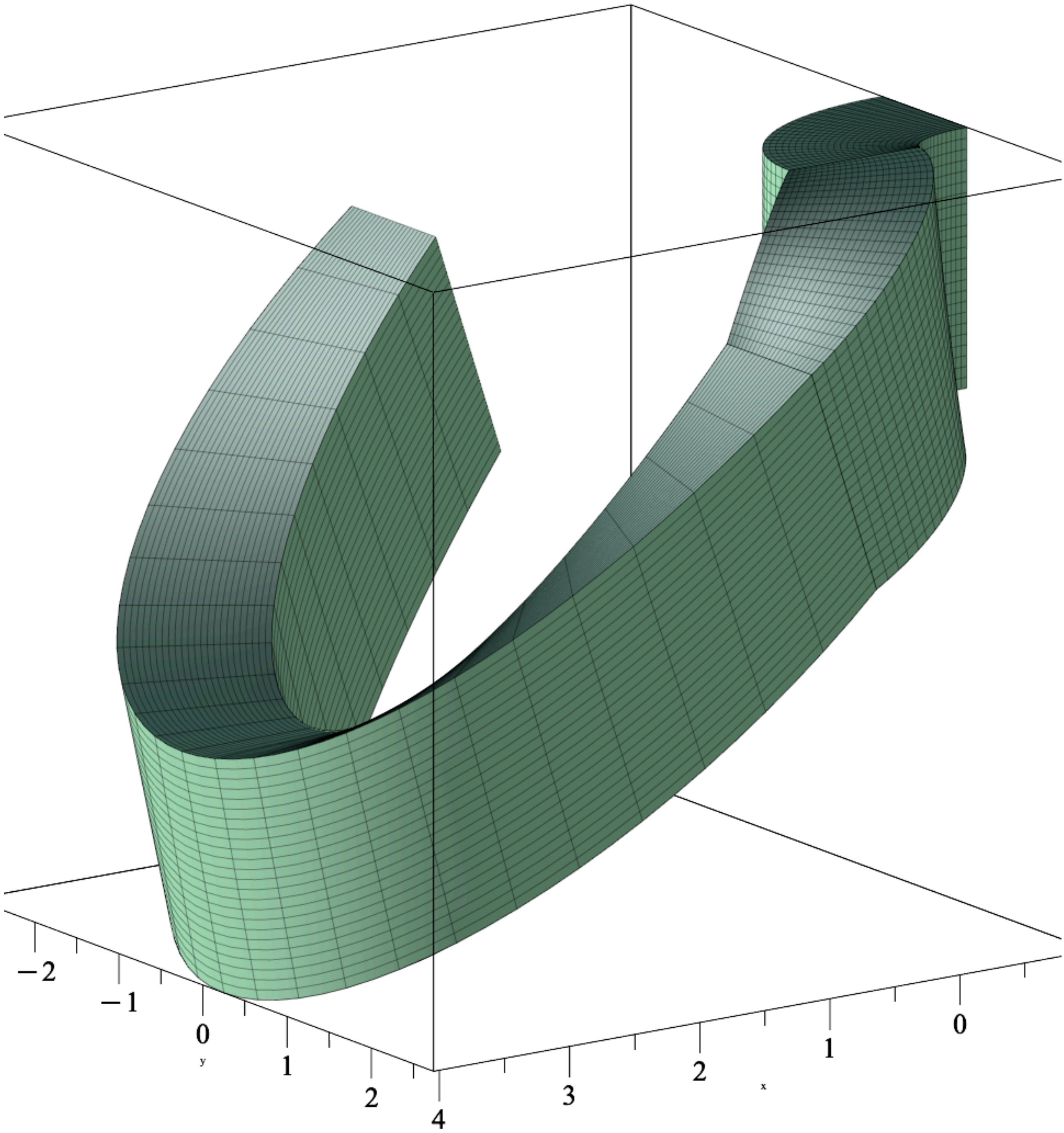
$$q33(t,u) = \left[-0.511500\sqrt{3}\,(-3\sin(t)-1)-0.500000\sin(t)\,u-0.250000\cos(t)^2,\; 2.226\cos(t)-\frac{\cos(t)\,u}{2}+0.250000\sin(t),\; -1.53450\sin(t)-0.944514\right] \quad \textbf{(4.4.4)}$$

```
> qp33:=plot3d(q33(t,u),t=0..Pi, u=-b...b,color=mg):
> q34:= (t,u) -> convert((R3(t).w4(u))^(%T),list)+cirk3rot(t):
  'q34(t,u)'=q34(t,u)
```

$$q34(t,u) = \left[-0.511500\sqrt{3}\,(-3\sin(t)-1) - 0.500000\sin(t)\,u + 0.250000\cos(t)^2,\right.$$ **(4.4.5)**

$$\left. 2.226\cos(t) - \frac{\cos(t)\,u}{2} - 0.250000\sin(t), -1.53450\sin(t) - 0.078486\right]$$

```
> qp34:=plot3d(q34(t,u),t=0..Pi, u=-b...b,color=mg):
> display(qp11,qp12,qp13,qp14,qp21,qp22,qp23,qp24,qp31,qp32,qp33,
  qp34, labels=["x","y","z"])
```



Here er eksempel på rotation med Frenet-Serret-baserne som rotationmatrice

```
> qF31:= (t,u) -> convert((FSB(t).w1(u))^(%T),list)+cirk3rot(t):
```

```
'q31(t,u)'=q31(t,u)
```

$$q31(t,u) = \left[-0.511500\sqrt{3}\,(-3\sin(t)-1) + 0.500000\sin(t) - 0.250000\cos(t)^2 u,\ 2.72600\cos(t) + 0.250000\sin(t)\,u,\ -1.53450\sin(t) - 0.511500 - 0.433014\,u\right] \quad \textbf{(4.4.6)}$$

```
> qpF31:=plot3d(qF31(t,u),t=0..Pi, u=-a...a,color=mg):
> qF32:= (t,u) -> convert((FSB(t).w2(u))^(%T),list)+cirk3rot(t):
  'q32(t,u)'=q32(t,u)
```

$$q32(t,u) = \left[-0.511500\sqrt{3}\,(-3\sin(t)-1) - 0.500000\sin(t) - 0.250000\cos(t)^2 u,\ 1.72600\cos(t) + 0.250000\sin(t)\,u,\ -1.53450\sin(t) - 0.511500 - 0.433014\,u\right] \quad \textbf{(4.4.7)}$$

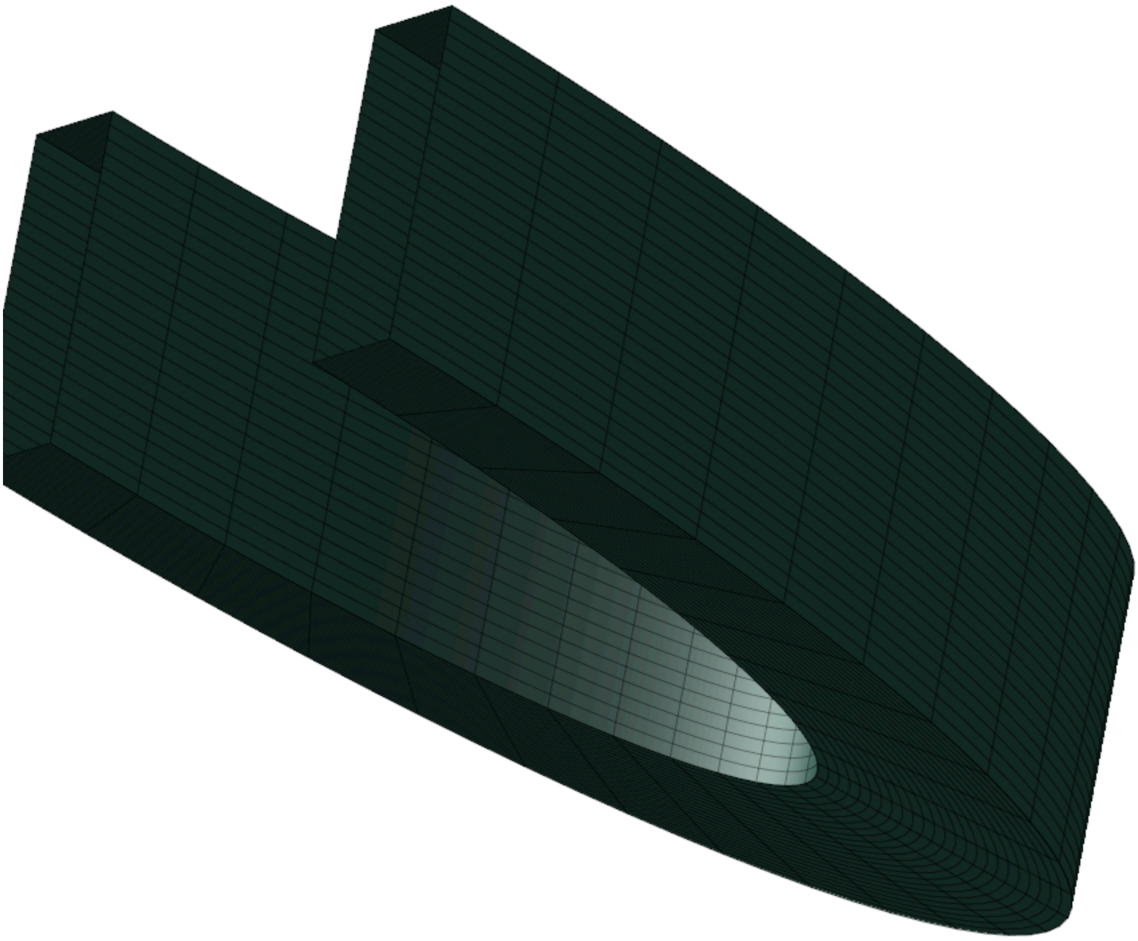
```
> qpF32:=plot3d(qF32(t,u),t=0..Pi, u=-a...a,color=mg):
> qF33:= (t,u) -> convert((FSB(t).w3(u))^(%T),list)+cirk3rot(t):
  'q33(t,u)'=q33(t,u)
```

$$q33(t,u) = \left[-0.511500\sqrt{3}\,(-3\sin(t)-1) - 0.500000\sin(t)\,u - 0.250000\cos(t)^2,\ 2.226\cos(t) - \frac{\cos(t)\,u}{2} + 0.250000\sin(t),\ -1.53450\sin(t) - 0.944514\right] \quad \textbf{(4.4.8)}$$

```
> qpF33:=plot3d(qF33(t,u),t=0..Pi, u=-b...b,color=mg):
> qF34:= (t,u) -> convert((FSB(t).w4(u))^(%T),list)+cirk3rot(t):
  'q34(t,u)'=q34(t,u)
```

$$q34(t,u) = \left[-0.511500\sqrt{3}\,(-3\sin(t)-1) - 0.500000\sin(t)\,u + 0.250000\cos(t)^2,\ 2.226\cos(t) - \frac{\cos(t)\,u}{2} - 0.250000\sin(t),\ -1.53450\sin(t) - 0.078486\right] \quad \textbf{(4.4.9)}$$

```
> qpF34:=plot3d(qF34(t,u),t=0..Pi, u=-b...b,color=mg):
> display(qpF31,qpF32,qpF33,qpF34, axes=none, labels=["x","y","z"]
  )
```



```
> display(qp31,qp32,qp33,qp34,axes=none, labels=["x","y","z"])
```